

Johan Sebastian Ospina Buitrago

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EXPERIENCE

Markforged

R&D + Senior Machine Learning Engineer

Oct 2022 – Present

Boston, MA

- Led cross-team delivery of the Markforged Layer Scan feature on FX10 printers, including orchestration, product planning, and architecture decisions across hardware, firmware, and software teams to support a system capable of handling 10s of requests per hour and responding within 5 minutes for end users for gigabyte scale image asynchronous input and output payloads using AWS, S3, Python, Typescript, and PostgreSQL.
- Built and scaled an image-to-3D model search system using DINOv3 embeddings and C++ filament renderer to deliver sub-500ms search results; owned model management, embedding generation, PostgreSQL pgvector storage compatible output, including embedding-search tooling
- Designed and operated long-running client/server workflows for internal materials-tuning analyses, improving system reliability and reducing validation effort from approximately two quarters to one month
- Adopted agentic workflows and helped automate internal tasks to improve team efficiency and reduce manual operational overhead via Context + Skill Creation
- Productionized ML image classification models using Amazon SageMaker, MLflow, and Hugging Face, and built AWS Lambda-based pipelines for composite image generation using computational photography and OpenCV
- Built and maintained data systems for terabyte-scale time-series telemetry and model evaluation workflows that accelerated material print tuning from months to weeks
- Mentored teammates and contributed to a patented 3D printing method in collaboration with engineering and science partners

PTC - Vuforia

Senior Prototyping Engineer

June 2021 – Sep 2022

Boston, MA

- Minimized positional drift from SLAM-to-Context-Graph data structure in Unity and C# Mobile frameworks and AngularJS/Lit Element Web Frontend using Graph Isomorphisms
- Reduced technical debt in production codebases by optimizing build tools, including configuring linters, formatters, and CI/CD pipelines for improved code quality and development efficiency in a team of 5 engineers

Princeton University

Research Assistant

Dec 2019 – May 2021

Princeton, NJ

- Developed a pipeline using Amazon Web Services, Internal Princeton Infrastructure, and Database systems to gather quality image correspondence data from globally distributed users
- Engineered a [Python Software Pipeline](#) 🔄 to reproduce [research](#) 📄 on calculating Camera Intrinsic Parameters using Image Correspondences and Projective Geometry

Wayfair

R&D Software Engineer II

Internship 2016 + June 2017 – March 2019

Boston, MA

- Led research into emerging immersive shopping technologies, exploring WebXR, ARKit, ARCore, Unity, BabylonJS, and automated material conversion from ray-traced assets to PBR workflows.
- Designed, implemented, and published [Doll House Projection Mapping Experience](#) 📄 at CHI EA '20 with a team of 3 employees
- Wrote OpenGL Shading Language (GLSL) GPU Vertex and Fragment Shader Code for Real Time Graphics Applications

EDUCATION

Princeton University

Master of Science in Computer Science

Sep 2019 – May 2021

Princeton, NJ

Boston University

Bachelor of Science in Computer Engineering, Magna Cum Laude

Sep 2013 – May 2017

Boston, MA

SKILLS

Programming Languages : Python, TypeScript, JavaScript, C#, $\LaTeX$, Java, Bash, C++, Swift, SQL

Libraries/Frameworks : Node.js, React, ESLint, Jest, Vitest, Redux, Mocha/Chai, PyTorch, NumPy, Pandas, Parquet, OpenCV, Plotly, MLflow, Scikit-Learn, Jupyter, Matplotlib, CUDA, Zustand

Tools/Platforms : Git, GitHub, VS Code, Cursor, Unix, Unity, AWS Lambda, Docker, CircleCI, AWS Athena, AWS ECR, AWS Kinesis, AWS SageMaker, Hugging Face, ClickHouse, Kubernetes, PostgreSQL, Datadog, Amazon Bedrock, Esbuild, Redis, Terraform

Domain Knowledge : Optimization Techniques, Linear Algebra, 3D Printing, Machine Learning, Neural Networks, Computer Vision, Computer Graphics, Computational Photography, Edge Computing, Getting Buy-In, Deep Learning, NLP, Time Series Analysis, Model Deployment, A/B Testing, MLOps, Statistical Analysis, Data Engineering, Distributed Computing

Languages : Spanish, French